

DHAMODARAN NARAYANA PERUMAL

Principal Mobility Lead & AppSec Specialist

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PROFESSIONAL SUMMARY

Visionary Mobile Application Engineering Lead and AppSec Specialist with 4+ years of experience architecting native, cross-platform, and hybrid mobile ecosystems while engineering deep-tech security mitigations. I specialize in owning the entire mobile lifecycle—from high-level solutioning, Clean Architecture design, and DevOps automation to proactive vulnerability research and RASP implementation. Recognized for exceptional analytical problem-solving and meticulous attention to detail when neutralizing complex threats (e.g., OWASP Mobile Top 10, dynamic instrumentation, root/jailbreak cloaking, SSL Pinning bypasses). An adaptable and continuous learner, I excel at leading cross-functional squads and communicating complex security concepts to stakeholders and engineering teams to deliver secure, production-ready, enterprise-grade deployments.

SKILLS & TECHNOLOGIES

- **Mobile Architecture & Cross-Platform:** MVVM, Clean Architecture, React Native, Kotlin Multiplatform (KMP), Flutter, Cordova, Capacitor, Ionic, Custom Plugin Architecture.
- **Native & UI Development:** SwiftUI, Jetpack Compose, Android OS Internals, Android NDK & JNI (C/C++), Java/Kotlin, Objective-C/Swift, Angular Material, Vanilla JS.
- **DevOps, CI/CD & Observability:** Fastlane, Jenkins, GitHub Actions, Git, App Store Connect (TestFlight), Google Play Console, Firebase Crashlytics, Dynatrace, AppDynamics.
- **Mobile AppSec & Proactive Research:** OWASP Mobile Top 10, Root/Jailbreak Detection & Evasion (Magisk/KernelSU), Runtime Tampering, SSL Pinning, VAPT Remediation, STRIDE Threat Modeling.
- **Security Tooling & Cryptography:** Frida, Ghidra, IDA Pro, Objection, Burp Suite, DexGuard, AppProtectt, Talsec, OAuth2, JWT, Google Play Integrity (Server-side validation).
- **Backend, Databases & Enterprise Middleware:** Event-Driven Architecture (EDA), Kafka, RabbitMQ, ActiveMQ, JMS, EJB, RMI, Redis, NoSQL (MongoDB, DynamoDB), Elasticsearch, VectorDB.
- **Core Competencies:** Technical Leadership & Mentoring, Global Stakeholder Management, Agile Delivery (Jira), Code Reviews, Analytical Problem-Solving, Effective Communication, Time Management.

PROFESSIONAL EXPERIENCE

Intellect Design Arena Ltd | Chennai, India

Consultant (Mobility Lead & AppSec Specialist) | Feb 2024 – Present

- **Architecture & Team Leadership:** Lead 3 cross-functional agile squads in the architecture, system design, and impact analysis of flagship mobile financial applications. Spearheaded transition strategies from legacy architectures to high-performance **React Native** and **Kotlin Multiplatform (KMP)** ecosystems.
- **DevOps & Automated Publishing:** Engineered and automated zero-touch mobile CI/CD pipelines leveraging **Fastlane** and Jenkins. Managed secure code signing, provisioning, and automated publishing of 6+ distinct enterprise client deployments to Apple TestFlight and Google Play Console.
- **VAPT Remediation & Threat Mitigation:** Spearheaded the remediation of 25+ critical and high-risk vulnerabilities following Gray Box VAPT audits (e.g., Yes Bank IGTB). Successfully mitigated complex authentication flaws, including 2FA/OTP bypasses and business logic exploitation.
- **Runtime Protection & Native Engineering:** Engineered robust defense-in-depth mechanisms utilizing **Android NDK & JNI (C/C++)** to block dynamic instrumentation (Frida, Objection) and detect advanced rooting (Magisk Hide, KernelSU). Integrated advanced app hardening frameworks (DexGuard, AppProtectt, Talsec) utilizing logic scattering to prevent reverse engineering.
- **Secure UI & Platform Integrity:** Prevented the "Janus" vulnerability and application repackaging by enforcing V2/V3 signature schemes. Enforced robust UI-level safety measures, including tapjacking defenses (filterTouchesWhenObscured) and screen-capture blocking (FLAG_SECURE).
- **Mentorship & CCB Leadership:** Mentored developers globally on secure coding standards. Authored attack/defense proof-of-concept (POC) scripts to demonstrate vulnerabilities during Change Control Board (CCB) reviews.

Capgemini | Chennai, India

Consultant | Jul 2023 – Sep 2023

- **Legacy API & Backend Hardening:** Managed, tested, and secured REST APIs for a major banking client. Evaluated legacy Enterprise Java backend services (utilizing **EJB and RMI**), patched critical logic vulnerabilities, and enforced secure data serialization between clients and servers to protect sensitive transactional payloads.

Intellect Design Arena Ltd | Chennai, India

Associate Consultant (Product Engineer - Mobility) | Jun 2022 – Jul 2023

- **Hybrid Mobile Optimization:** Built and maintained responsive, hybrid banking applications. Leveraged deep Vanilla JavaScript (ES6+), HTML5, and CSS3 expertise to resolve rendering bottlenecks and optimize them on memory-constrained devices.

- **Vulnerability Assessment:** Intercepted API traffic and performed packet analysis using **Burp Suite** to identify, document, and patch critical data leakage points, insecure session cookies, and unauthorized client-side access vectors.

Otomate Technologies | Chennai, India

Full Stack Developer | Aug 2021 – Mar 2022

- **Full-Stack Architecture:** Designed secure, versioned REST APIs utilizing robust password-hashing algorithms and secure OAuth token lifecycles. Integrated high-performance data layers utilizing **MongoDB, DynamoDB, and Redis** for caching.
- **Infrastructure Management:** Managed AWS cloud environments with a focus on secure networking (VPCs), IAM roles, and scalable deployments.

CKS Solutions | Nagercoil, India

Web Development Intern | Nov 2018 – May 2019

- **Feature Contribution:** Collaborated with engineers to develop modular web components and optimize SQL queries for analytics workloads.

KEY RESEARCH & SECURITY PROJECTS

- **Aran RASP (Mobile Security SDK):** Spearheaded the R&D and implementation of an enterprise-grade Runtime Application Self-Protection (RASP) SDK.
 - *Research:* Utilized reverse engineering tools (**Frida, Ghidra, IDA Pro**) to analyze and bypass root, jailbreak, and runtime-hooking evasion techniques.
 - *Development:* Built low-level security wrappers in C/C++ (JNI) to intercept runtime tampering, stop active debugging, and prevent network spoofing. Applied native-level obfuscation and logic scattering to resist static reverse engineering.
- **Corporate Banking Mobile Platform (CMS - eMACH.ai):** Served as lead architect and security owner for an enterprise-level banking suite operating on an Event-Driven Architecture (EDA). Led the frontend migration to modern Capacitor/React Native implementations. Seamlessly integrated mobile gateways with the backend **Apache Kafka** event backbone and **ActiveMQ/JMS** message queues to ensure real-time, highly concurrent financial transaction processing while resolving high-severity VAPT findings.

EDUCATION & AWARDS

- **Bachelor of Engineering (Computer Science)** | Stella Mary's College of Engineering, Nagercoil, India | 2018 – 2021 (First Class)
- **SPOTLIGHT (GEM) Award (2025):** Recognized for outstanding technical leadership and analytical problem-solving, clearing critical iOS/Android build and security failures to unblock multi-platform production launches.
- **SPOT Award (2024):** Received direct client appreciation for exceptional technical debugging, AppSec vulnerability mitigation, and cross-platform UI resolution.